

EARTH, ENVIRONMENTAL, AND PLANETARY SCIENCES

Some Meteorite Information

The chemical composition of lunar soil

As a lunar geochemist I have been approached many times by people who believe that they have a sample from the moon. Common stories are (something like) “This dust was given to my late grandfather by astronaut Buzz Lightyear” or “This rock that I found in my petunia pot looks just like lunar meteorite [QUE 94281](https://sites.wustl.edu/meteoritesite/items/lm_que_94281/) (https://sites.wustl.edu/meteoritesite/items/lm_que_94281/) on your website.” Lately, people have been sending me reports that they have obtained by chemical analysis from commercial laboratories or a [hand-held x-ray analyzer](#). (<https://sites.wustl.edu/meteoritesite/items/meteorite-testing/#XRF>) So, here is what you need to know in order to interpret those reports.

Major Elements – In lunar rocks and soils 98–99% of the mass consists of 7 chemical elements

Oxygen (41-45%) | [Silicon \(Si\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Si) (<https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Si>) | [Aluminum \(Al\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Al) (<https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Al>) | [Calcium \(Ca\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Ca) (<https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Ca>) | [Iron \(Fe\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Fe) (<https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Fe>) | [Magnesium \(Mg\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Mg) (<https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Mg>) | [Titanium \(Ti\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Ti) (<https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Ti>)

with the following ratios

[Fe/Mn](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#FeMn) (<https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#FeMn>) | [Ca/Al](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#CaAl) (<https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#CaAl>)

[composition-of-lunar-soil/#CaAl](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#CaAl) | [Mg# \(https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#MgFe\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#MgFe)

Minor Elements – Nearly all of the remaining 1% consists of only 4 chemical elements

[Manganese \(Mn\) \(https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Mn\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Mn) | [Sodium \(Na\) \(https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Na\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#Na) | [Potassium \(K\) \(https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#K\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#K) | [Phosphorus \(P\) \(https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#P\)](https://sites.wustl.edu/meteoritesite/items/the-chemical-composition-of-lunar-soil/#P)

Below are charts I have made from data from dozens of literature sources and my own lab for what geochemists call the “major elements” and “minor elements” in samples from the 6 Apollo, 3 Russian Luna, and 2 Chinese Chang’e missions that have brought soil samples back from the moon. To make it simple, I have stuck to just soil (regolith) samples, not rocks. I have also included data for those lunar meteorites that are breccias because many to most of these rocks are composed of lithified soil. The lunar meteorites come from all over the moon whereas the Apollo and Luna mission all come a small area of the nearside.

In rocks of the Earth and Moon, oxygen is the most abundant chemical element, 41-45% on the moon. Practically nobody actually measures the concentration of oxygen in rocks anymore. We measure the “metals” like iron and aluminum.

Terrestrial geochemists like to “express” the measured concentration of, say, silicon “as the oxide.” They measure the concentration of Si and state the concentration as percent SiO₂. So, [10.0 % Si is 21.4% SiO₂ \(http://meteorites.wustl.edu/goodstuff/oxides.htm\)](http://meteorites.wustl.edu/goodstuff/oxides.htm). Quartz is a form SiO₂, but quartz is rare on the moon. Nearly all (>99%) lunar Si is in the silicate minerals plagioclase, pyroxene, and olivine, but not quartz. Likewise, there is no actual MgO (the mineral periclase) on the moon; magnesium is carried mostly by the minerals pyroxene and olivine. We express the metal concentrations as oxide concentrations because the sum of the 10 major and minor metal oxides should be 100±1%. If not, something wrong with the analysis (!) as there are no (= insignificant amounts of) carbonates, sulfates, or hydrous (water-bearing) minerals on the moon. Lunar

meteorites, however, often contain carbonates, sulfates, or hydrous minerals as a result of weathering on Earth after they land.

So, for geochemists, the bottom and left axes of the plots below are in weight-percent oxide. For scrap-yard dealers and jewelers who might have a handheld x-ray analyzer set to the “metal” setting, use the top and right axes.

All the plots have aluminum concentrations on the horizontal axis. I do it that way because Al varies over a large range in lunar samples. (To confuse you, [elsewhere here](https://sites.wustl.edu/meteoritesite/items/how-do-we-know-that-its-a-rock-from-the-moon/) (<https://sites.wustl.edu/meteoritesite/items/how-do-we-know-that-its-a-rock-from-the-moon/>)) I have put FeO+MgO on the horizontal axis, but that is OK because there is a strong anticorrelation (<https://sites.wustl.edu/meteoritesite/items/how-do-we-know-that-its-a-rock-from-the-moon/#Anticorrelation>) between Al_2O_3 and FeO+MgO in lunar samples.) In all the plots, points with high Al_2O_3 concentrations represent soils from the feldspathic highlands (light-colored regions of the moon) and those with low Al_2O_3 are from the basaltic mare (dark circular regions).

Finally, in the plots below, each point for Apollo 11, the 3 Luna missions, and the 2 Chang’e (CE) missions represents a soil sample from a single location. For example, nearly all the Apollo 11 points represent soil sample [10084](https://curator.jsc.nasa.gov/lunar/lsc/10084.pdf) (<https://curator.jsc.nasa.gov/lunar/lsc/10084.pdf>), which is probably the most well characterized geologic sample ever analyzed. For Apollos 12, 14, 15, 16, and 17, each point represents a numbered soil sample (“surface” and “trench” soils, no cores, collected along traverses, e.g., samples 12032, 14163, 15071, 65701, and 76501 (mean of all available analyses for each)). The large spread for some of these missions reflects the compositional variation among the various locations at which samples were collected at the site. For the lunar meteorites, each point represents a named stone, e.g., MacAlpine Hills 88105 or Northwest Africa 032. For reference, each plot also includes an “Earth” point, which is an average of four different estimates I found in the literature for the mean composition of upper continental crust of the Earth.

Silicon (Si)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

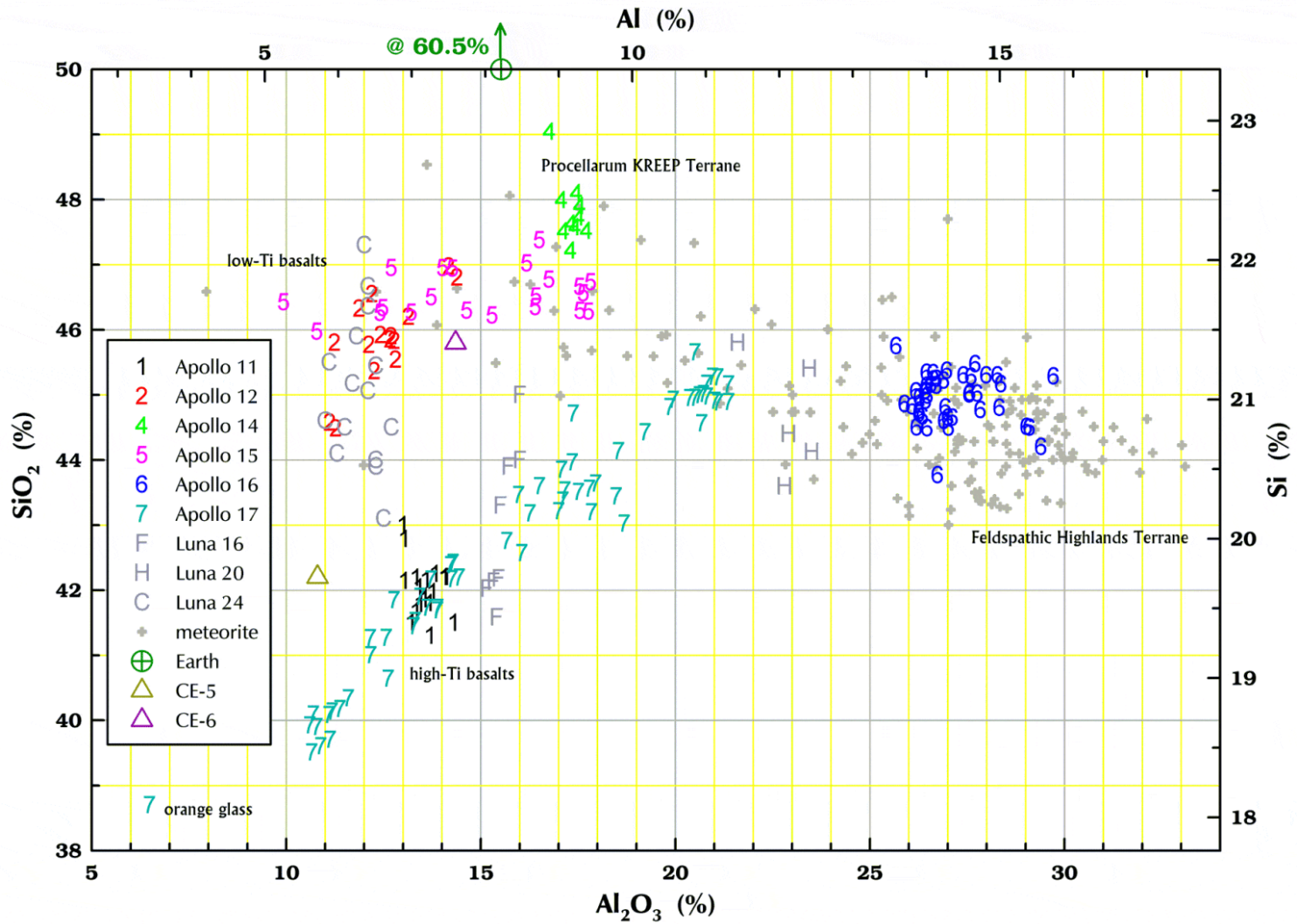


Figure 1. On Earth, SiO_2 concentrations in rocks vary from 0% to 100%. The variation on the moon is much less because the 3 major minerals in lunar rocks, plagioclase feldspar (usually anorthite), pyroxene, and olivine, all have about the same SiO_2 concentration.

Iron (Fe)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

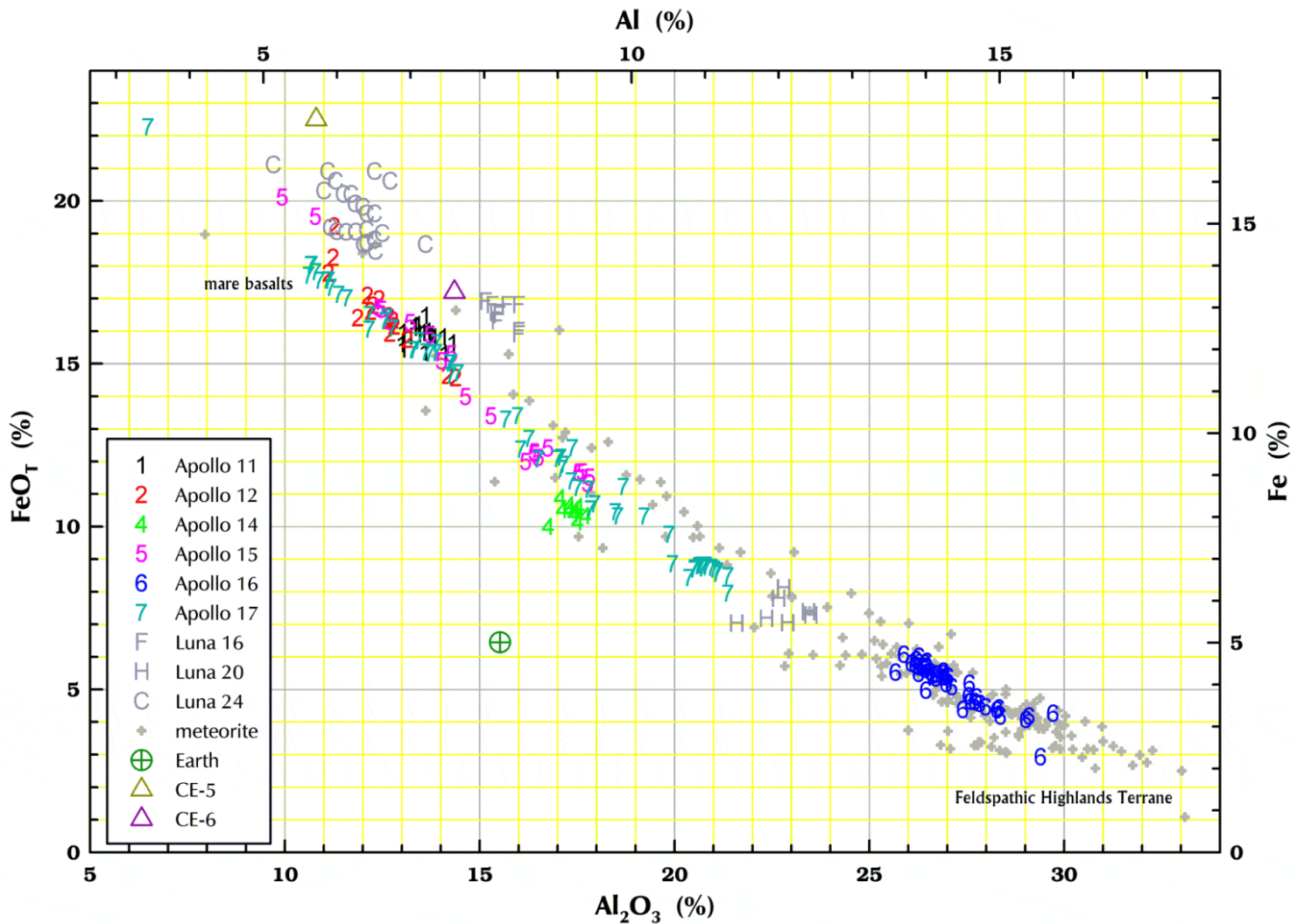


Figure 2. On Earth, iron exists in the 2+ (ferrous) and 3+ (ferric) oxidation states so in chemical analysis of terrestrial rocks, Fe concentrations are usually stated as % Fe_2O_3 because on Earth the ferric oxidation state is more common than ferrous oxidation state. On the moon there is effectively no oxygen-bearing atmosphere so there is effectively no iron 3+ iron minerals. The iron in pyroxene, olivine, and iron-titanium minerals like ilmenite is all in the ferrous (2+) oxidation state. To complicate the issue, some of the iron in every lunar soil exists as metal. Up to 10% of the iron in some of these sample is metallic, usually as iron-nickel metal derived from meteorites. So, in analyses of lunar samples, results for iron are usually stated as “total Fe as FeO” or FeO_T . The anticorrelation in this plot occurs because soils on the left (basaltic) are dominated by the Al-poor, Fe-rich minerals pyroxene, olivine, and ilmenite whereas those on the right (feldspathic) are dominated by the Al-rich, Fe-poor mineral plagioclase.

Manganese (Mn)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

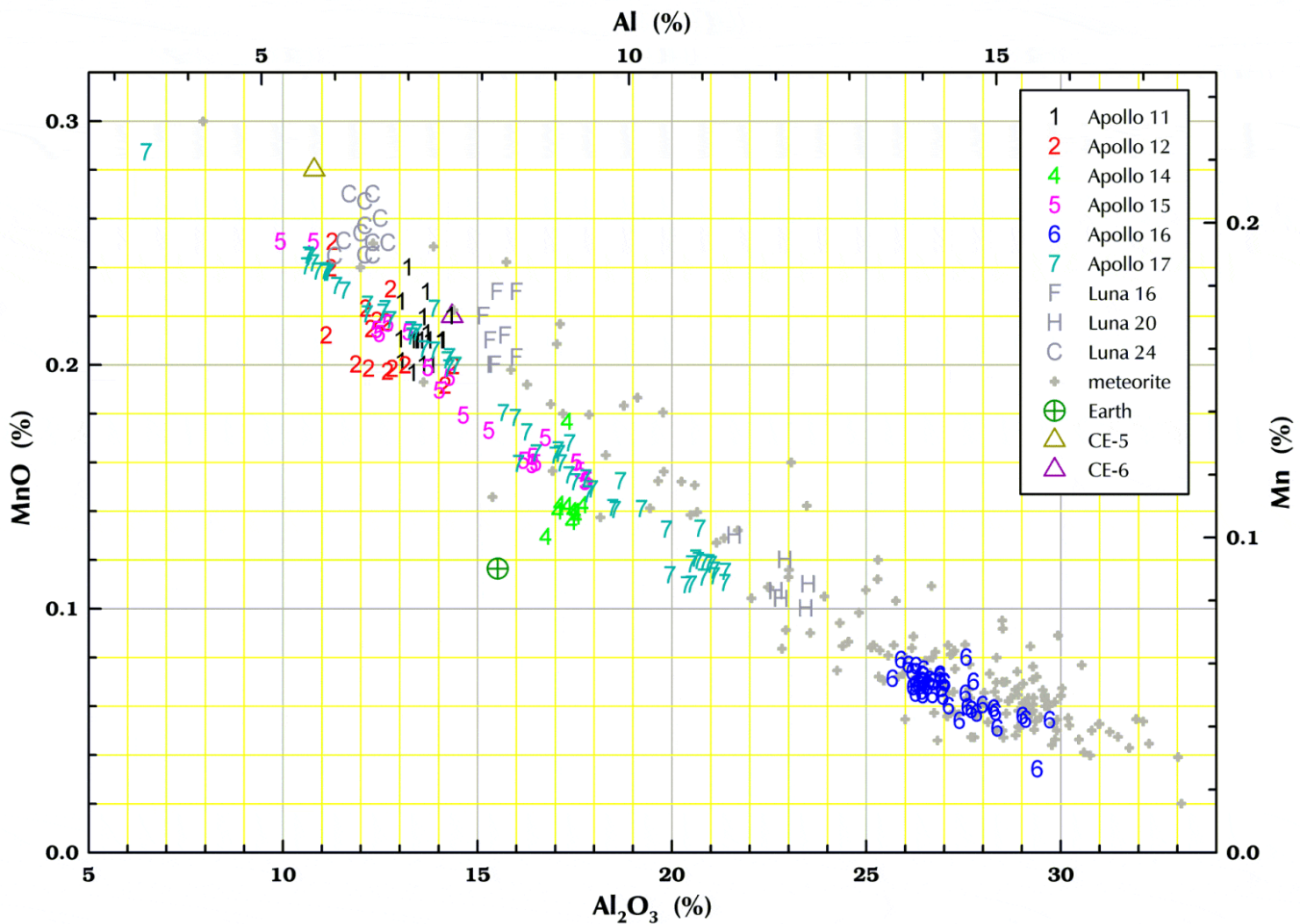


Figure 3. On the moon, all the Mn is in the 2+ oxidation state and Mn^{2+} has essentially the same ionic size as Fe^{2+} . As a consequence, Mn “behaves” just like Fe, leading to a strong correlation.

Iron/Manganese (Fe/Mn)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

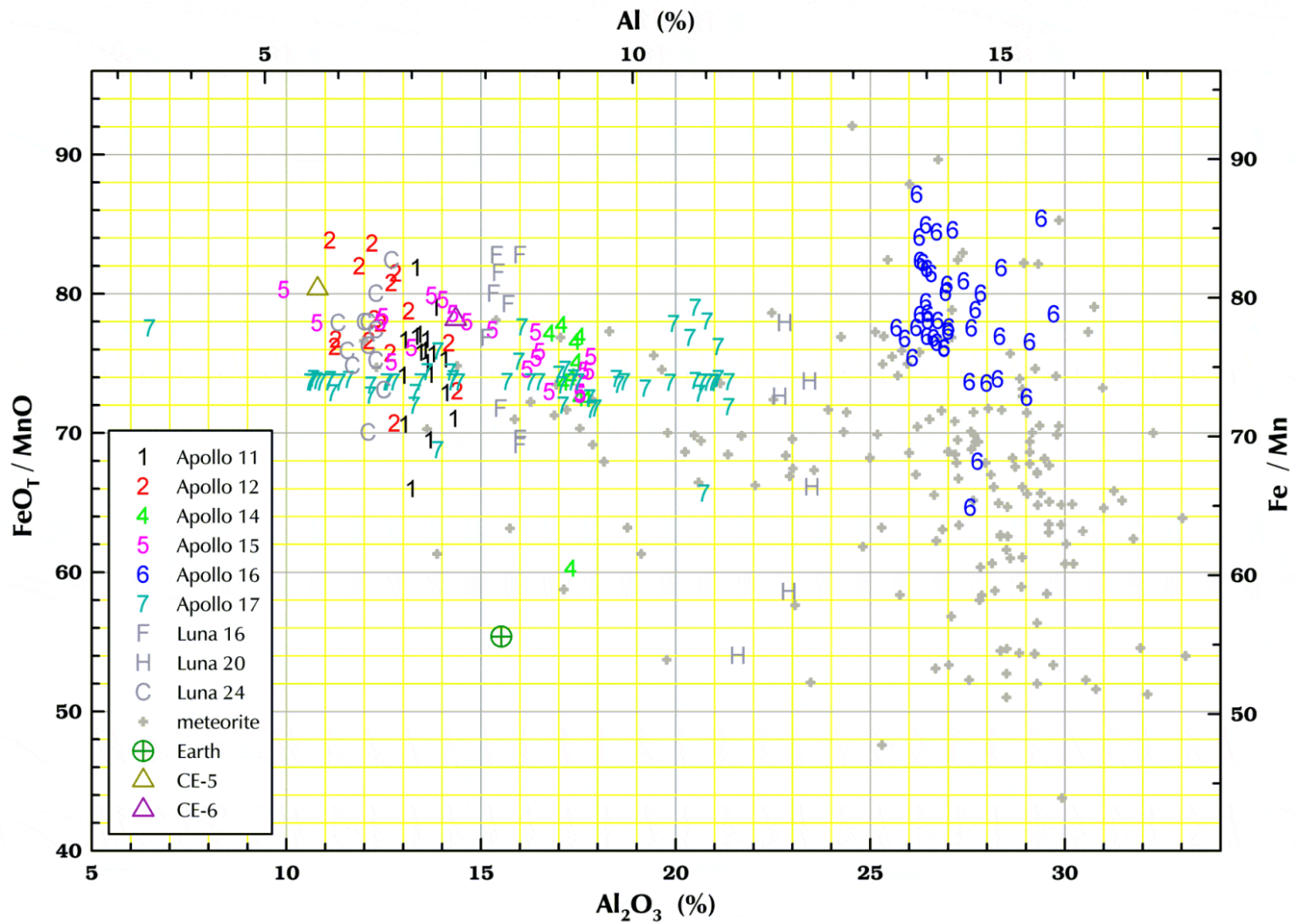


Figure 4. As a result of the similarity in Mn and Fe behavior, Fe/Mn ratios in bulk samples of lunar soils and rocks are rather constant in the 60-90 range. This characteristic is useful for distinguishing lunar meteorites from other types of meteorites.

Magnesium (Mg)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

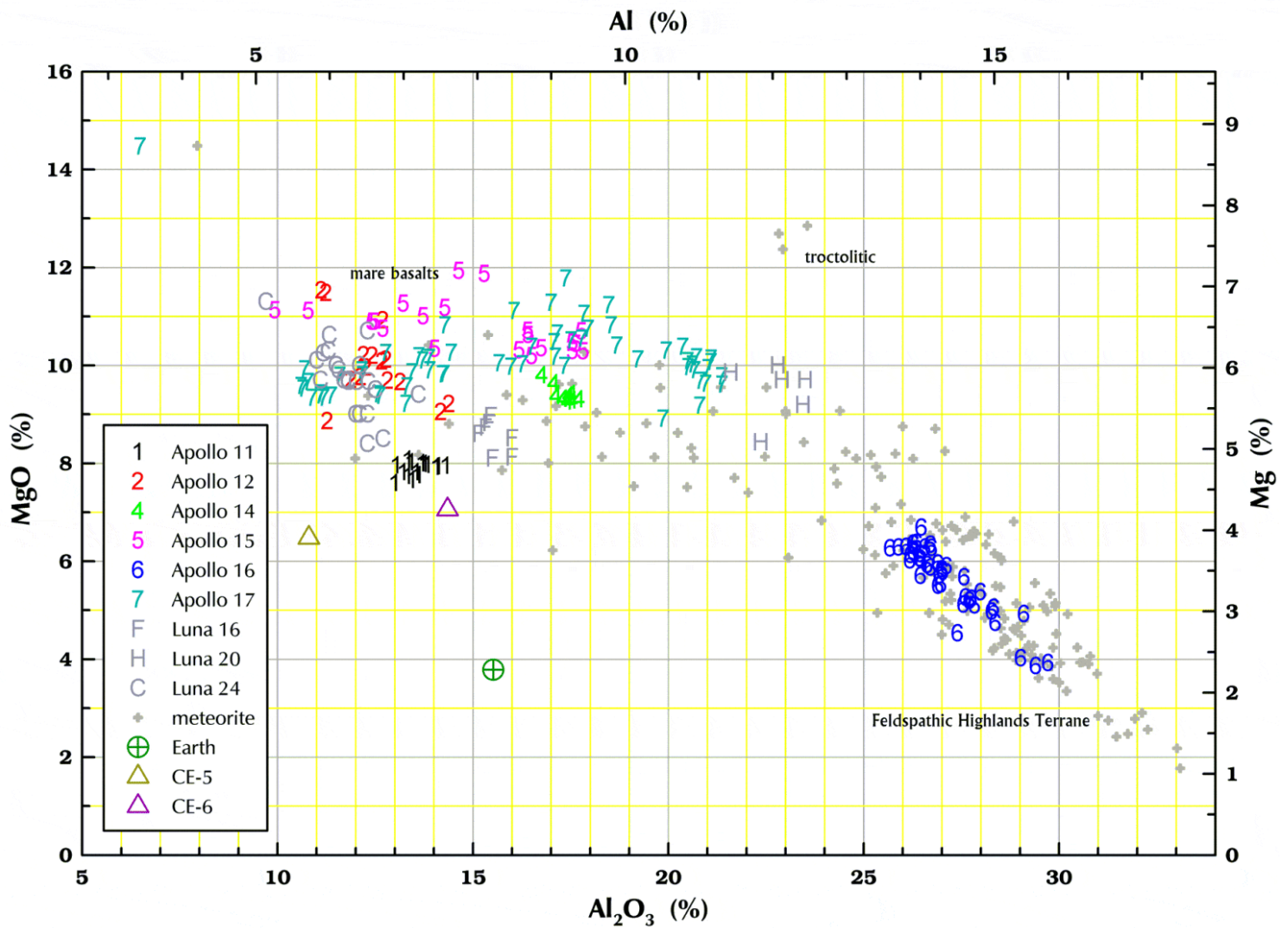


Figure 5. Most of what is said above for 2+ Fe is also true for magnesium. In lunar rocks, nearly all the Mg is in pyroxene and olivine. Note that both Chang'e-5 and Chang'e-6 soils have low MgO compared to Apollo basalts.

“Magnesium Number”

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

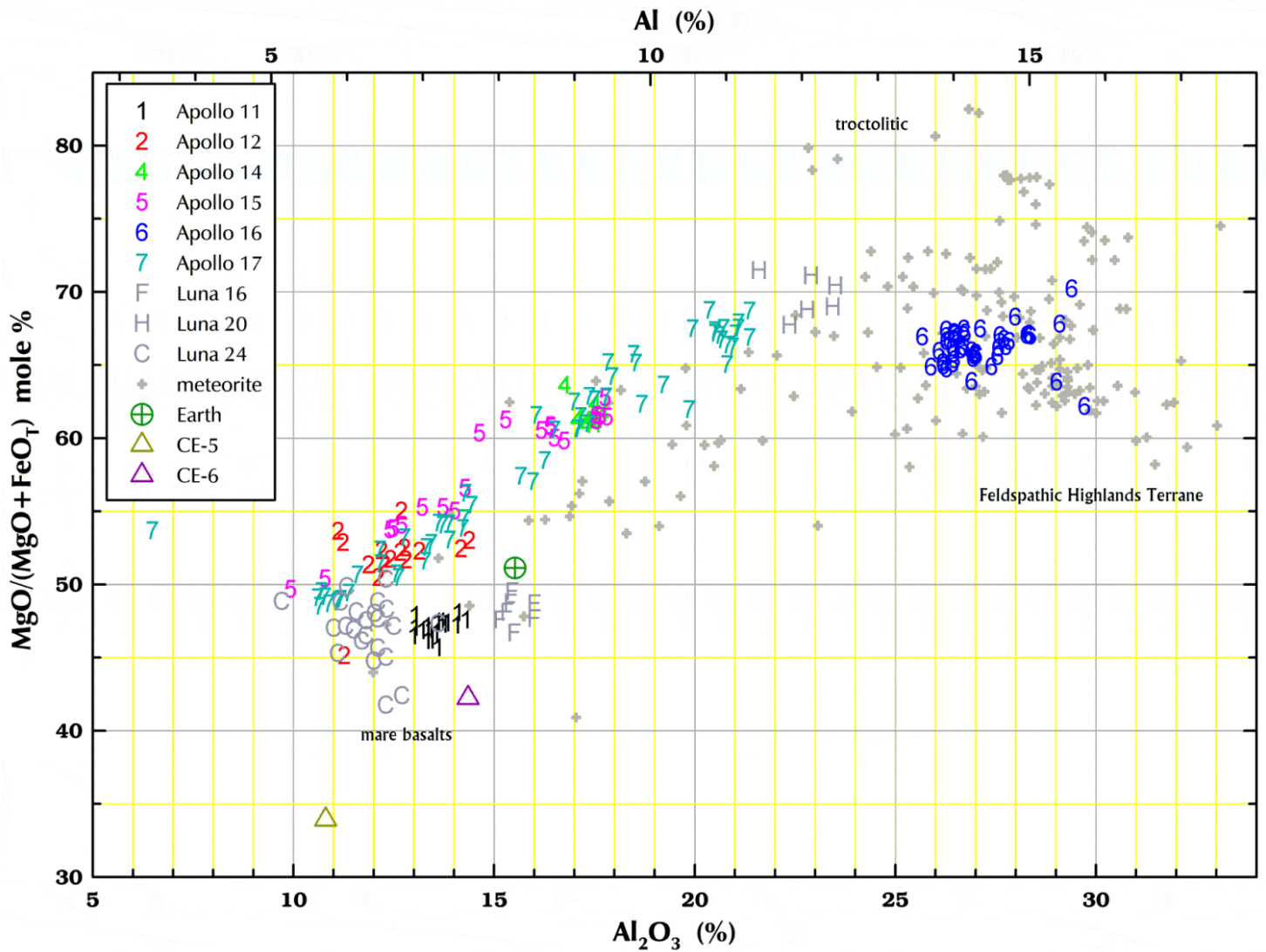


Figure 6. The “magnesium number” (bulk mole percent $\text{MgO}/[\text{MgO} + \text{FeO}_T]$) is usually lower in mare basalts than rocks and soils from the feldspathic highlands or the PKT.

Calcium (Ca)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

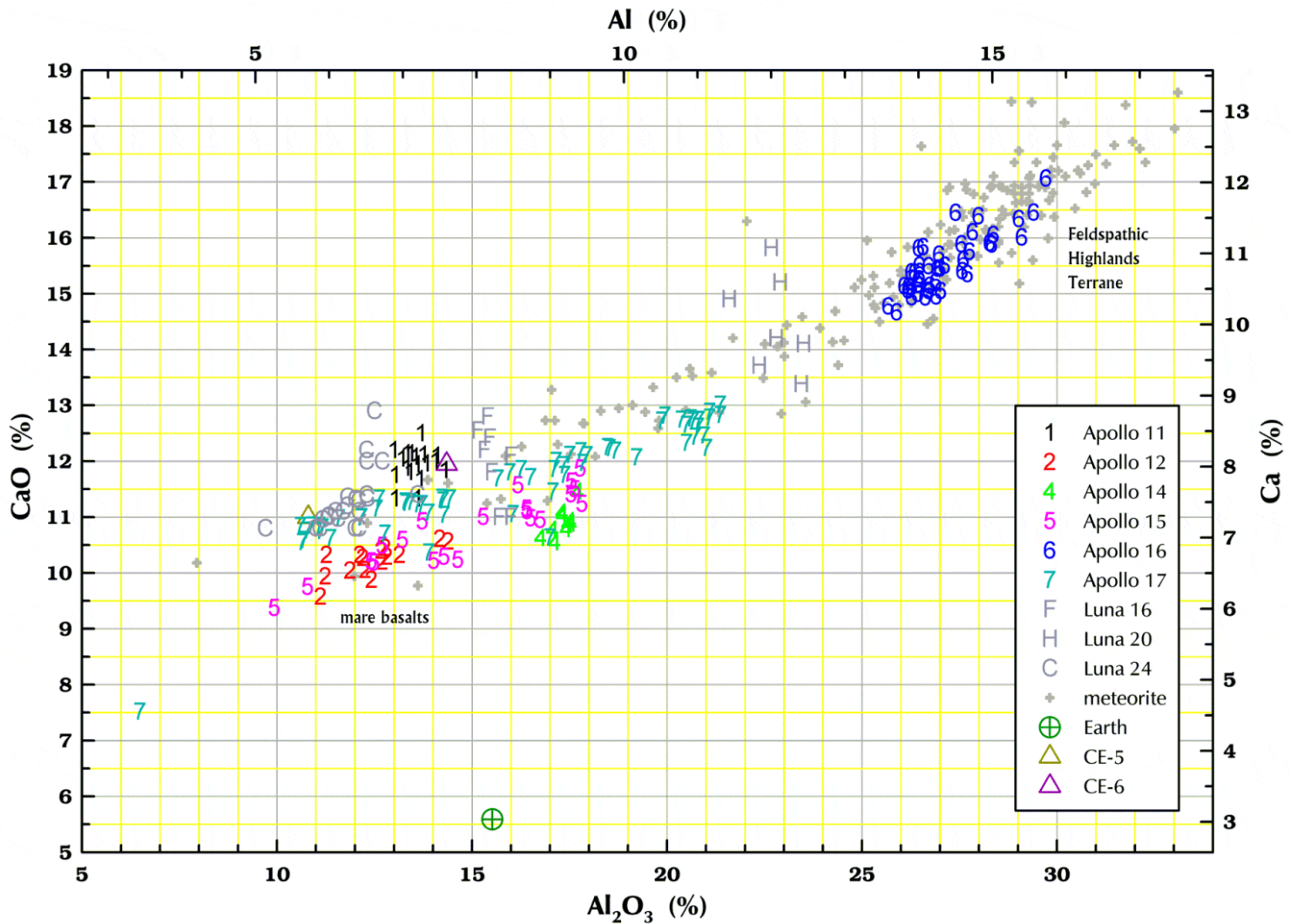


Figure 7. In the feldspathic highlands, nearly all of the Ca is in plagioclase, as is the Al, so the two elements strongly correlate. For Al-poor lunar rocks, e.g., mare basalts, some of the Ca is in pyroxene.

Calcium/Aluminum (Ca/Al)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

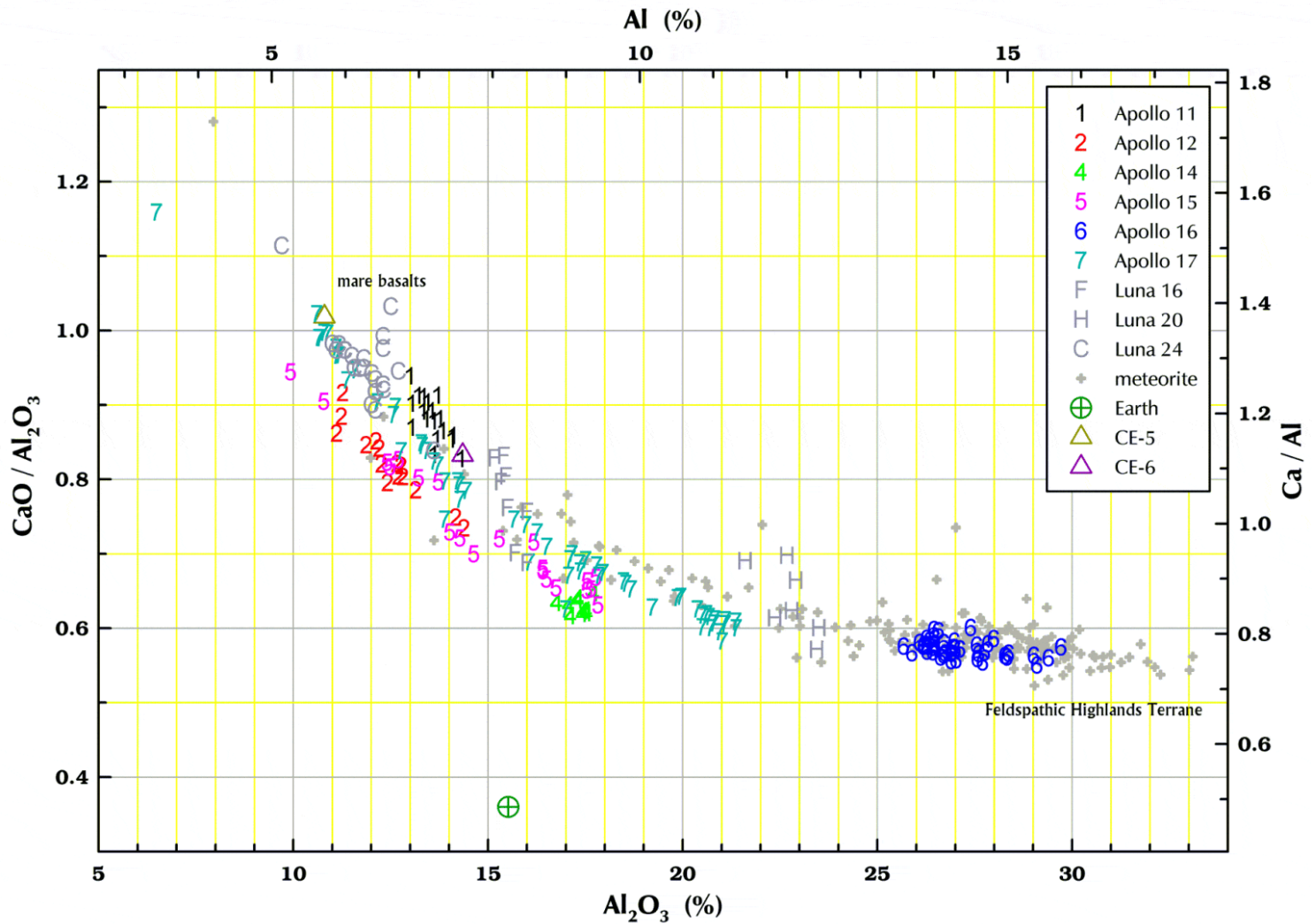


Figure 8. The Ca/Al ratio in lunar samples varies by only a factor of 2. In the high Al_2O_3 samples, both elements are contained almost exclusively in plagioclase so the ratio is constant. In the low Al_2O_3 samples (basaltic soils) some Ca is contained in pyroxene. High-Ca meteorites are contaminated with terrestrial calcite.

Titanium (Ti)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

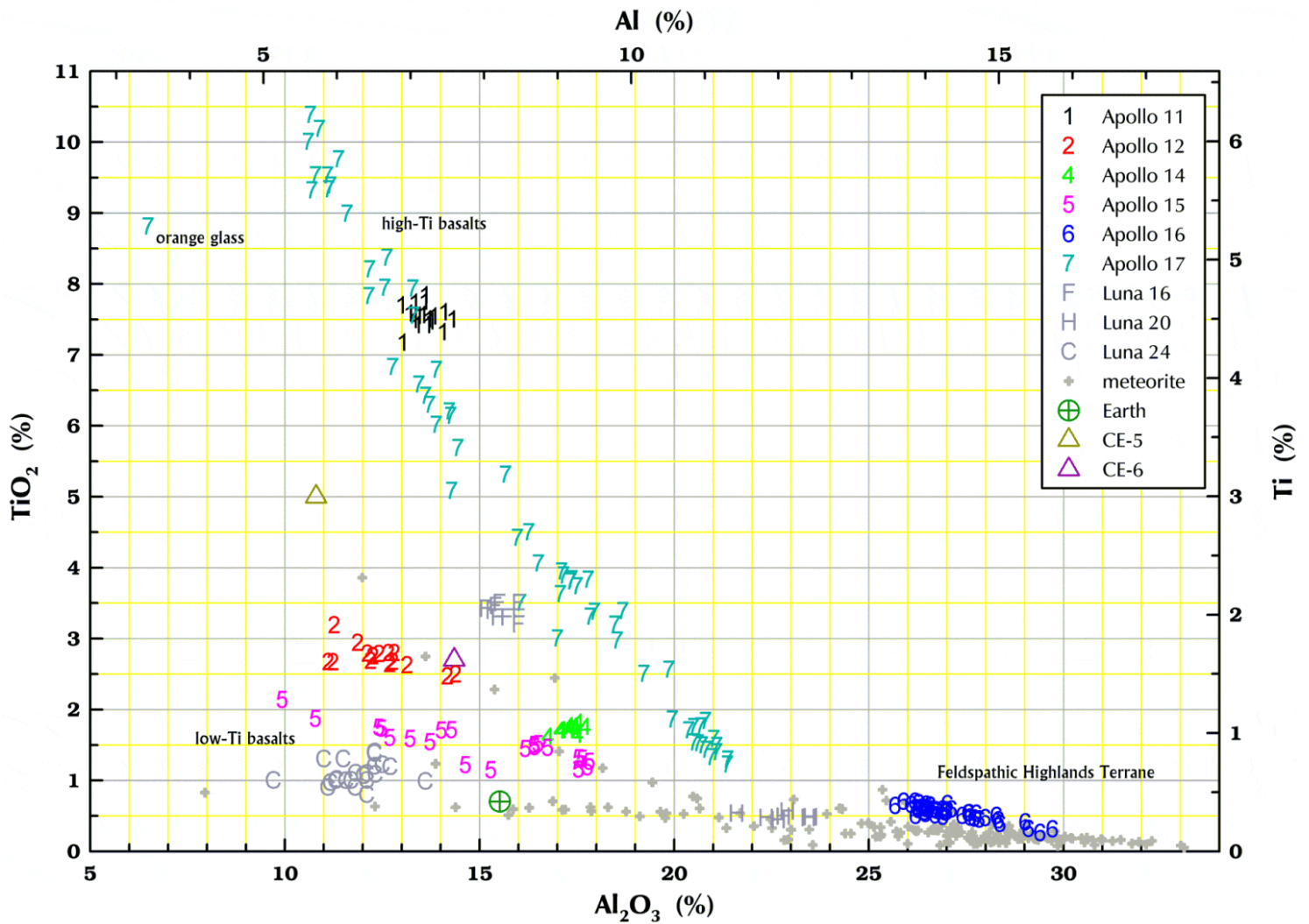


Figure 9. Concentrations of Ti vary by a factor of 10 in lunar basalts and basaltic lunar soils. Ti is the most highly variable major-element on the moon.

Chromium (Cr)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

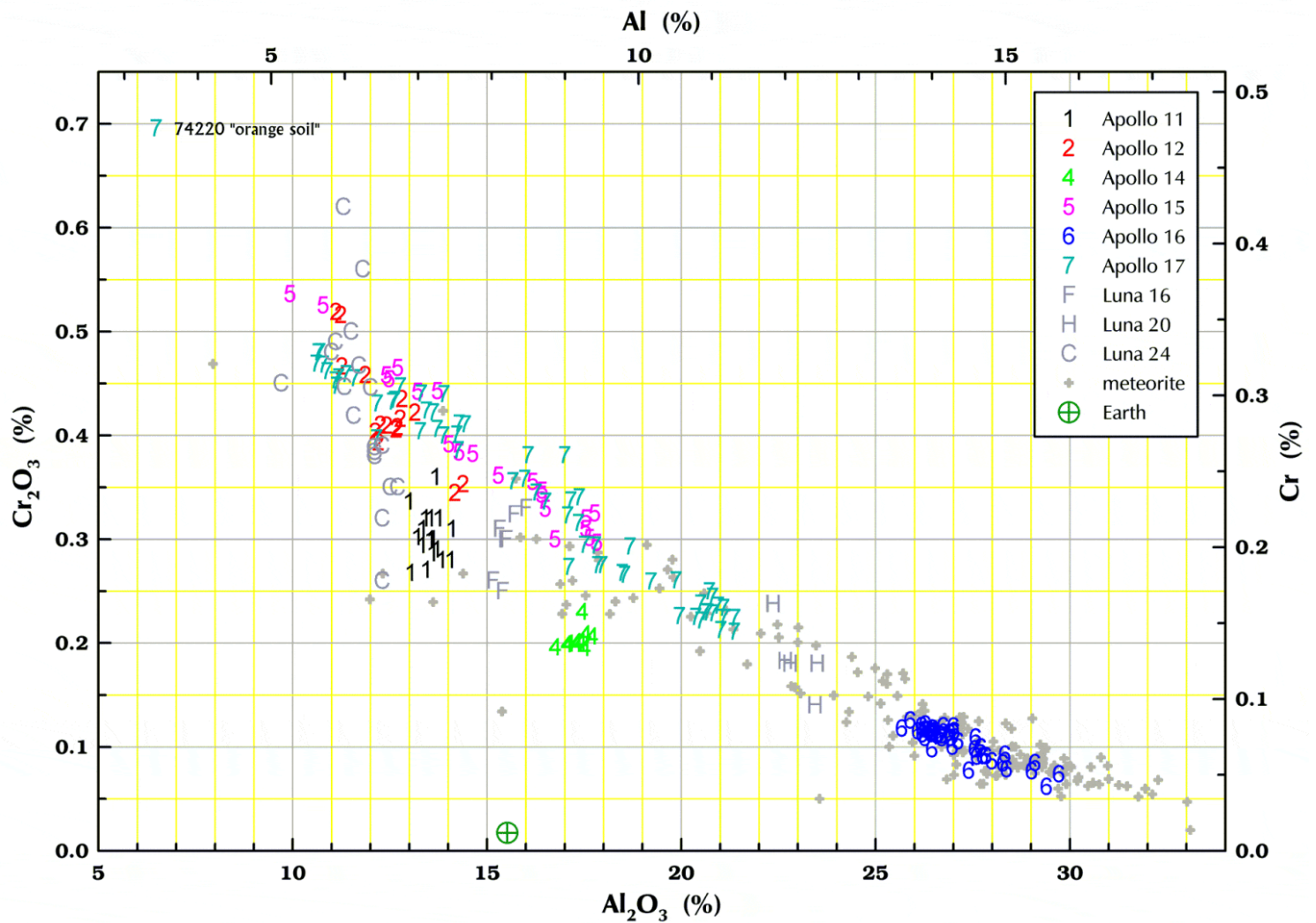


Figure 10. Cr concentrations in lunar samples are much greater than they are in nearly all terrestrial rocks. Cr is one of the best elements for distinguishing between lunar meteorites and terrestrial rocks. I have not found Cr data for the Chang'e samples.

Sodium (Na)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

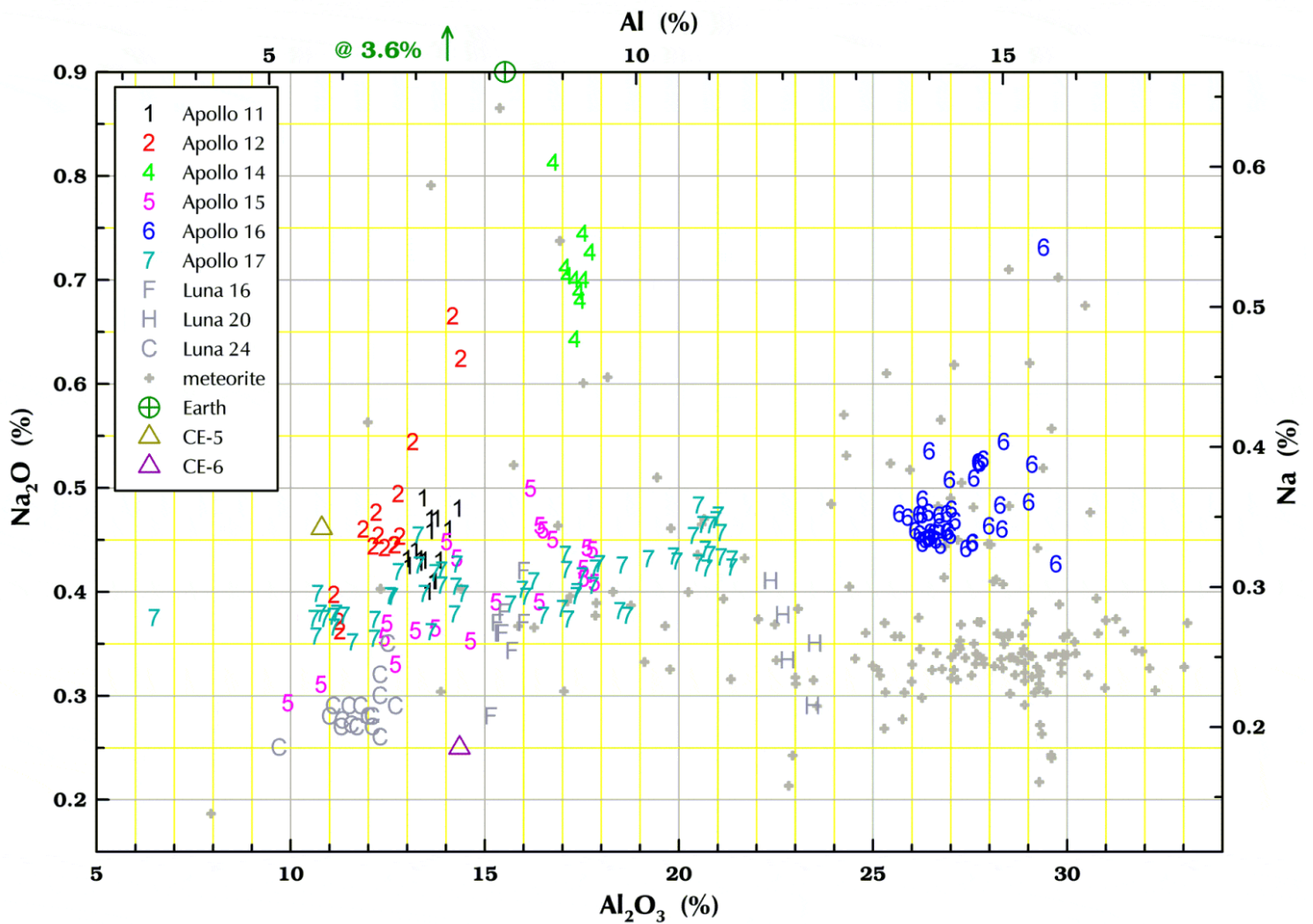


Figure 11. Na concentrations in lunar rocks and soils are much lower than they are in most terrestrial rocks. Na is an element that is often good for distinguishing between lunar meteorites and terrestrial rocks.

Potassium (K)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

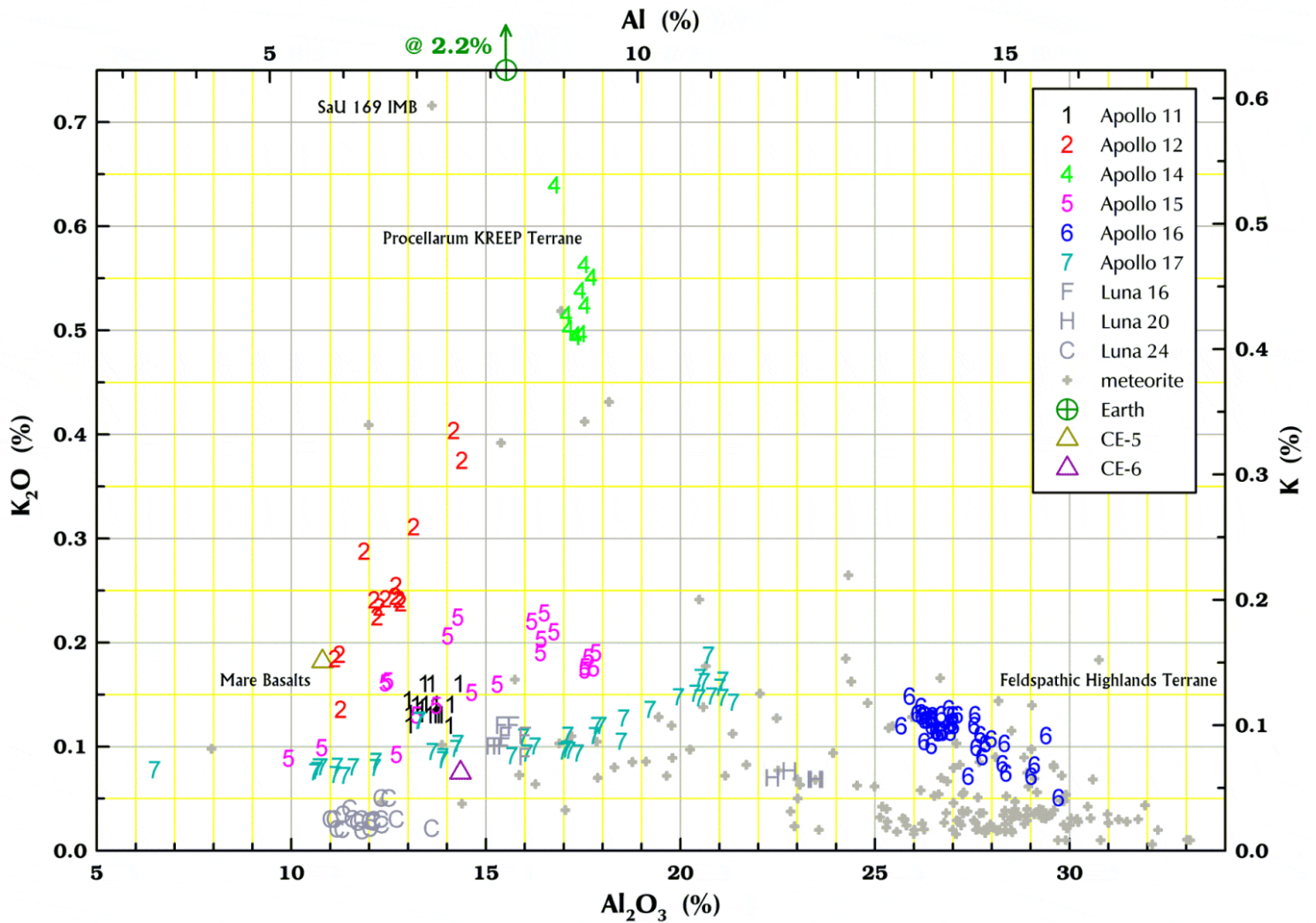


Figure 12. Like Na, K concentrations in lunar rocks and soils are much lower than they are in most terrestrial rocks. Potassium is an element that is often also good for distinguishing between lunar meteorites and terrestrial rocks.

Phosphorus (P)

Compositions of Soils from the Apollo, Luna, and Chang'E Missions and Brecciated Lunar Meteorites

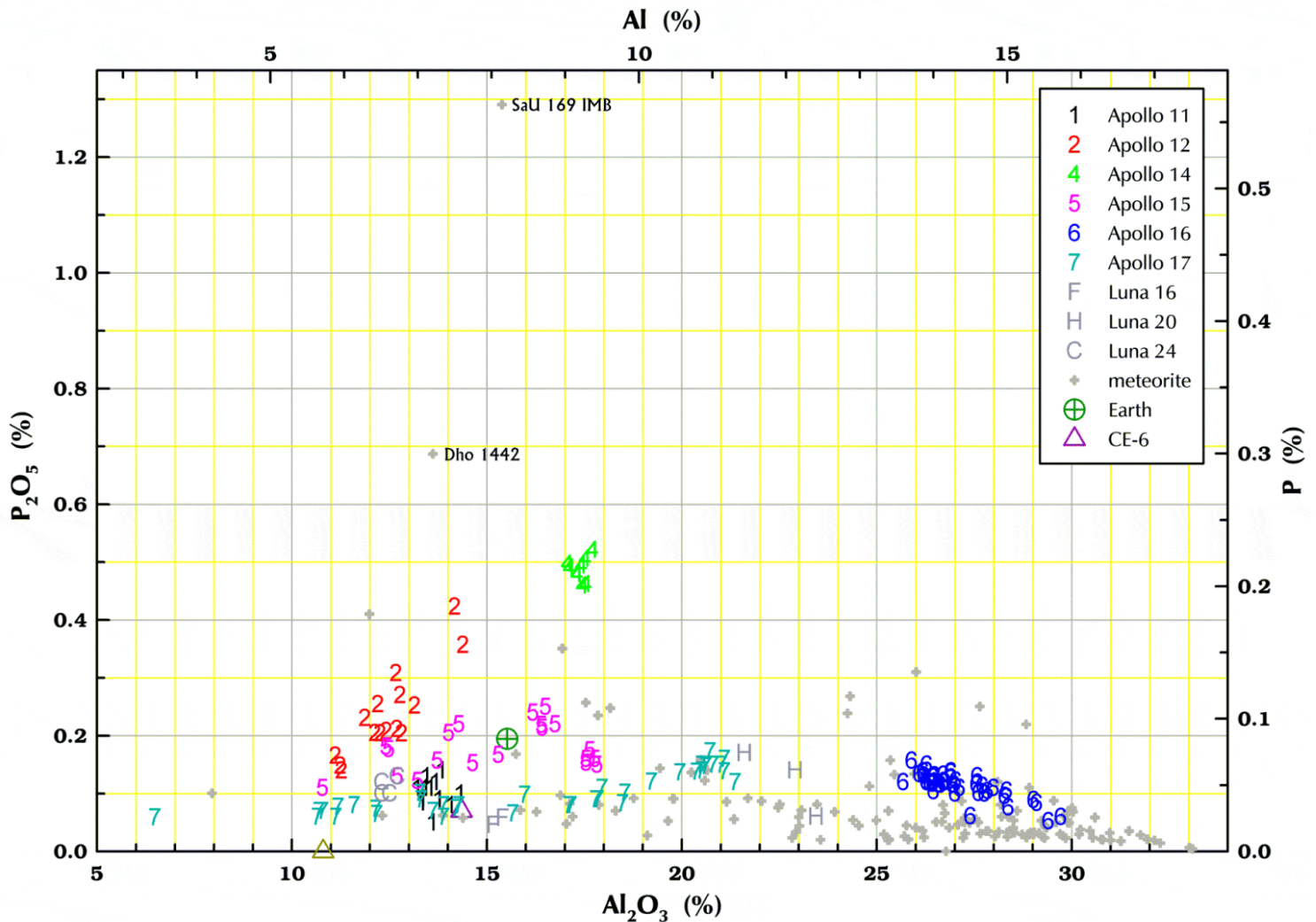


Figure 13. Concentrations of K, P, and Na are high in Apollo 14 soils, some soils from Apollo 12, and a few lunar meteorites because they sampled a geochemically anomalous region of the moon known as the Procellarum KREEP Terrane or PKT. Minor and trace elements that are incompatible with the crystal structures of plagioclase, pyroxene, and olivine are high in the PKT and occur mainly in minor and trace minerals like potassium feldspar (K), apatite (rare earth elements [REE] and P), and zircon.

Sources of Chang'E data

Li Chunlai, Hu Hao, Yang Meng-Fei, Pei Zhao-Yu, Zhou Qin, Ren Xin, Liu Bin, Liu Dawei, Xingguo Zeng, Zhang Guangliang, Zhang Hongbo, Liu Jianjun, Wang Qiong, Deng Xiangjin, Xiao Caijin, Yao Yonggang, Xue Dingshuai, Zuo Wei, Su Yan, Wen Weibin and Ouyang Ziyuan (2022) Characteristics of the lunar samples returned by the Chang'E-5 mission. *National Science Review* 9: nwab188.

Li Chunlai, Hu Hao, Yang Meng-Fei, Liu Jianjun, Zhou Qin, Ren Xin, Liu Bin, Liu Dawei, Zeng Xingguo, Zuo Wei, Zhang Guangliang, Zhang Hongbo, Yang Saihong, Wang Qiong, Deng

Xiangjin, Gao Xingye, Su Yan, Wen Weibin, Ouyang Ziyuan (2024) Nature of the lunar farside samples returned by the 5 Chang'E-6 mission. *National Science Review* 11: nwae328.

Lunar Meteorites

Lunar Meteorite Compositions

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